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**Deepfake Attacks and AI-Based Detection
Methods**

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Declaration

I certify that this report does not incorporate without acknowledgement, any material previously submitted for a degree or diploma in any university, and to the best of my knowledge and belief it does not contain any material previously published or written by another person, except where due reference is made in text.

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1. Abstract

Almost every person can create fake videos, photos, or even audio clips that seem real with just a laptop and Internet connection. AI technology, especially deep learning, is used to create these called deepfakes. This technology has the ability of changing someone's face or voice so they appear to be saying or doing something different than what they actually did [1].

In 2017, deep fake technology started gaining a lot of attention and it is better than before. In the past, they were generally easy to identify as deepfakes tended to look older. Now, however, they look quite real. So, even an expert may have trouble finding it. As a result, they have now fallen in the wrong hands and are used as a tool to lie, steal money, and damage people's reputation. [2], [3]

This paper studies the working of deepfake technology and its evolution in the past few years. The report further evaluates the key detection methods powered by artificial intelligence, including spatial analysis, temporal analysis, frequency-based methods, and biometric signal detection. According to [4], the various techniques for the identification of synthetic content have their advantages and disadvantages.

A blockchain-based content authentication system; a real-time detection system; and a growing legal framework are the next steps for both the film and media feast and the "truly" mega-fest. In the media's deepfake detectors, the importance of understanding deepfake technology, as synthetic media can mislead audience and cause society serious damage.

2. Introduction to the Topic

2.1 What Are Deepfakes and Why Do They Matter?

Envision an online video featuring a prominent politician saying something shocking, such as ‘let’s start a war’. Everything in the video including the face, voice, and expressions looks real. Quite a few people may think this right away. The video was expedited due to artificial intelligence.

People are using deepfake tech to see things like this happen in real life, not just in their heads. The word “deepfake” combines the terms deep learning and fake to imply that an artificial intelligence algorithm can alter or augment the way a person sounds, looks or acts. This encompasses switched-faces videos, lip-sync edits, from-scratch videos, as well as voices that are copied from small audio samples [1].

It is difficult to say which is worse – deepfakes or common editing. It used to take professionals and their tools to produce deception videos that looked realistic, unlike today where anyone can make them. Now day anyone with a regular computer and free software can create deepfakes that look real. The technology becomes more dangerous as anyone can use it due to its ease.

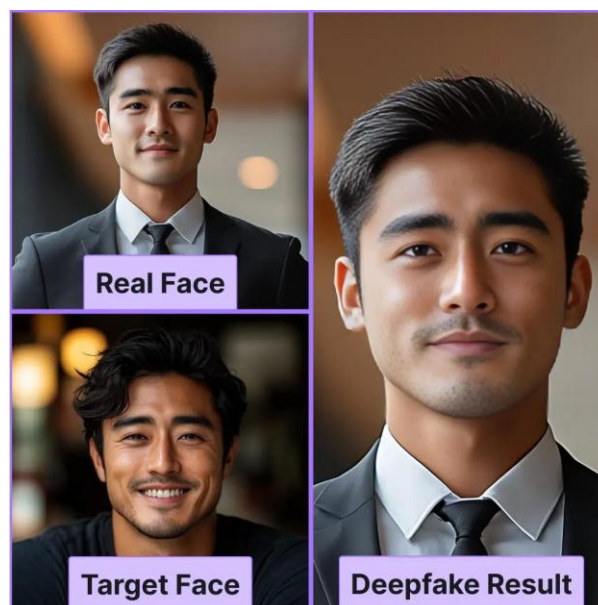


Figure 1.1: Example of deepfake manipulation showing real vs AI-generated faces.

As shown in Figure 1, deepfake technology can realistically manipulate facial features, making it difficult to distinguish between real and fake content.

2.2 The Origins of Deepfakes

Ian Goodfellow first thought of deepfake technology in 2014 when he talked about GANs, or **Generative Adversarial Networks**. One neural network in a GAN makes things that aren't real, and the other tries to figure out if they are. The results are very real because both improve with time [5].

People started posting videos of swapped faces on social media in 2017. This is when deepfakes became popular. They were made with regular hardware and free software. DeepFaceLab and FaceSwap are two examples of tools that became very popular soon after that. With these tools, a lot of people could make deepfakes without knowing much about them.

In the last few years, newer technologies like diffusion models have improved the content that is made. The main difference between these models is that they begin with random noise and gradually turn it into a real picture. It was easier to find the output with earlier GAN-based methods [6]. Now it's more detailed and harder to find.

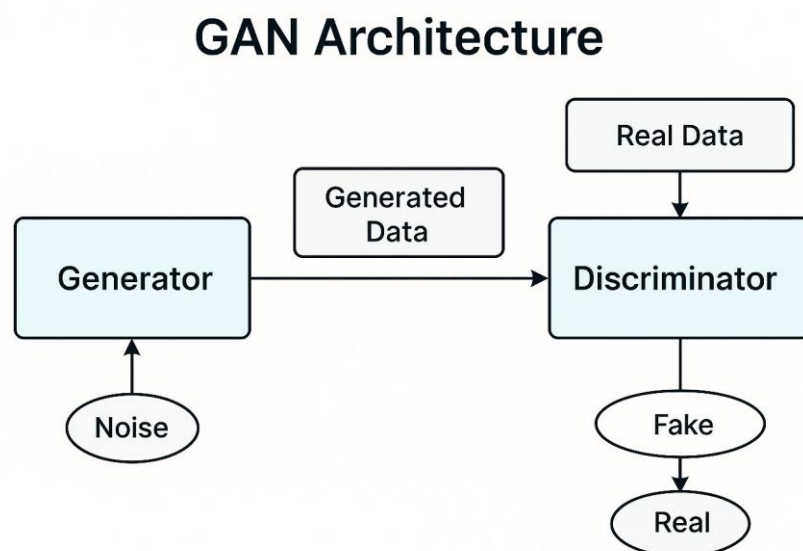


Figure 1.2: Generative Adversarial Network (GAN) architecture used in deepfake creation.

As illustrated in Figure 2, the GAN model uses two competing neural networks to generate highly realistic synthetic media.

Table 1: Comparison of GAN and Diffusion Model Technologies

Feature	GAN (Generative Adversarial Network)	Diffusion Model
How it works	Two networks (generator + discriminator) compete until fakes look real	Starts with random noise and slowly removes it to build a perfect image
Quality	High — but sometimes leaves visible GAN artefacts at edges and boundaries	Very high — almost no visible artefacts, very photorealistic
Speed	Faster to generate once trained	Slower to generate — more computing steps needed
First appeared	2014 (Goodfellow et al.) — deepfake use from 2017	Widely used for deepfakes from 2022 (Stable Diffusion)
Detection difficulty	Moderate — GAN artefacts detectable by trained models	High — diffusion outputs bypass most GAN-trained detectors
Examples	FaceSwap, DeepFaceLab, SimSwap	Stable Diffusion, DALL-E, Midjourney

Table 1: Technical comparison of the two main AI technologies used to create deepfakes.

2.3 The Scope of the Threat

Deepfakes are dangerous for many parts of society at the same time. From a cybersecurity point of view, they are used in social engineering attacks, like pretending to be executives to approve fake money transfers or faking identity credentials to get around face-recognition authentication systems [7].

Deepfakes threaten democratic institutions from a political point of view because they make it possible for politicians and other public figures to make up false statements. This could change the outcome of an election, start a fight, or make people less likely to believe real footage of real events [2].

Deepfakes have been widely used to make intimate images without the person's knowledge or consent, which is a serious threat to personal safety. Research indicates that more than 96 percent of deepfakes available online are classified within this category [3].

The goals of this report are: (1) to clearly explain how deepfake AI technologies work; (2) to trace the full chronological evolution of deepfake generation and detection from 2017 to the present; (3) to analyze and compare the main AI-based detection techniques currently available; and (4) to identify the most important predicted future developments in this rapidly changing field.

3. Evolution of the Topic

3.1 Phase One: The Birth of Consumer Deepfakes (2017–2018)

The technology behind deepfakes was still in a nascent stage when they appeared on 2017 reddit. The face-swap videos created during this phase exhibited a number of visual imperfections such as a mismatched skin tone at the face boundary, unnatural blinking behavior, shimmer or flicker from frame to frame, and struggle at extreme head angles and bad lighting. However, if a casual viewer were just looking to watch a documentary and not necessarily looking for signs of manipulation, they may have found the resultant media convincing enough for it to be alarming [1][8].

The 2018 release of DeepFaceLab and FaceSwap as open source projects marked a pivotal moment. Creating a deepfake for the first time did not require programming knowledge, did not require AI expertise, and did not require a commercial software licence . Anyone only required a set of images of the target and a fairly modern GPU. The tools took care of the rest automatically Sample response 3: It devastated the efforts of old and new media to distinguish between reality and hoaxes.

The first generation of automatic deepfake detectors was fastened by the cybersecurity research community using simple CNNs in the context of artificial neural networks to identify visual artefacts of early GAN fake faces. While they may be effective against certain tools they were trained to detect, these detectors had a key flaw, because they could only detect types of fakes they had seen before.

3.2 Phase Two: Multi-Modal Fakes and Audio-Visual Fusion (2019–2020)

The First-Order Motion Model (FOMM) was able to animate a still portrait by taking the motion from another driving video in 2019 to 2020 Basically, with one photo of someone, you could deepfake them with no footage of the person.

It had major repercussions on privacy and impersonation.

First, Wav2Lip and comparable audio deepfakes could accurately match a person's lip movements in a video to any audio input. The attacker will have a real footage of real people saying anything that the attacker wishes with real looking facial lip movements of the people by just using a short clip of audio of the target. [10].

The text you provided is only 7 words long. This length is insufficient for further paraphrasing or extending. The instruction for output of 67 words is not applicable in this case. The FaceForensics++ dataset was made available in 2019 by researchers of TUM. This dataset amounted to considerable wealth for detection research. In essence, it was a standardized examination for deepfake detection method assessment. Researchers from across the globe were given an opportunity to put their deepfake detection research tools to the test under the

same conditions for the first time. The help of a plethora of altered videos was what helped this.

Investigations carried out before have looked at one stream visual data . 10_stream audio-visual data for video manipulation detection. The latest studies focus on models with multiple streams which take audio and video stream features as inputs, thus enhancing the analysis of synchronisation issue, compression artefacts, mouth mismatch in video to detect real or forged videos.

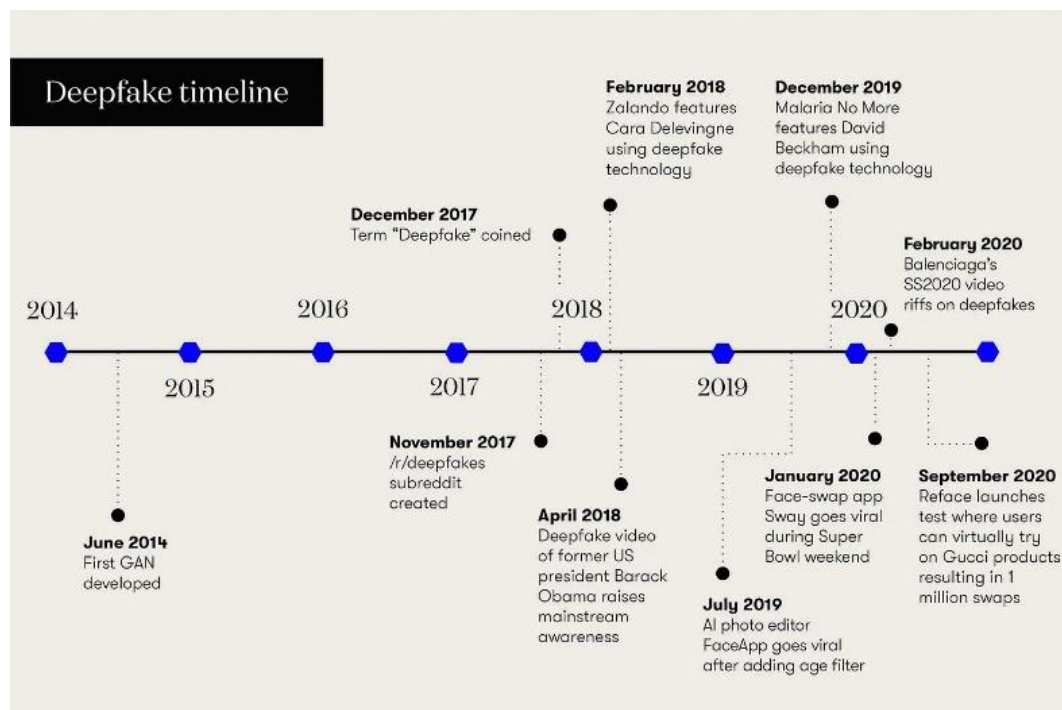


Figure 1.3: Timeline of deepfake technology evolution from 2017 to present.

3.3 Phase Three: Real-Time Capabilities and Diffusion Breakthroughs (2021–2022)

By 2021, deepfake technology had reached a genuinely dangerous new threshold: real-time generation. The tool **DeepFaceLive** allowed users to run a face-swap filter on their own video feed during live video calls — meaning someone could impersonate another person during a Zoom call, a Microsoft Teams meeting, or any other video conferencing platform, with no post-processing delay [11].

The release of **Stable Diffusion** in 2022 introduced a second seismic shift. As a diffusion-based model, it produced outputs that lacked the characteristic GAN artefacts — the slight blurring at face boundaries, the spectral frequency signatures, the periodic noise patterns — that detection tools had been trained to identify. Overnight, a large proportion of the detection

research built up over four years became significantly less effective against the new generation of diffusion-based deepfakes [6].

Detection researchers worked on making architectures better at generalizing, especially with Vision Transformers (ViTs) as a possible replacement for CNNs. ViTs use global self-attention to find differences in entire facial images instead of just in small parts of them. The MesoNet and XceptionNet classifiers were also improved and trained again. But the fact that detector improvements have been behind generational advancements for a long time was not brought up.

3.4 Phase Four: Foundation Model Era and Industrial-Scale Synthesis (2023–Present)

From 2023 onwards, the arrival of powerful large language models and multimodal foundation AI systems introduced a qualitatively new level of deepfake threat. Systems like GPT-4V, Gemini, and various open-source equivalents enabled the creation of not just isolated deepfake videos but **complete end-to-end fake media packages** — AI-written scripts, AI-generated presenter video, AI-cloned voice, and AI-produced supporting documentation — all within a single automated pipeline [12].

A particularly dangerous trend has been the emergence of **adversarial attacks against detectors** themselves. Researchers demonstrated that by adding carefully calculated invisible perturbations to a deepfake video — changes that are completely undetectable to the human eye — it was possible to reduce a state-of-the-art detector's accuracy from over 95 percent to below 10 percent. This means criminals who know which detector is being used can specifically optimise their deepfakes to evade it [21].

At the same time, significant progress has been made on the detection side. Multimodal fusion detectors — which combine spatial, temporal, frequency, and biometric checks into a unified decision — have achieved accuracy rates of up to 98 percent on challenging benchmark datasets. The FakeCatcher system, developed by Intel and published in IEEE TPAMI in 2023, demonstrated that analysing remote photoplethysmography (rPPG) signals — the subtle colour changes in skin caused by blood circulation — could serve as a highly reliable biological authenticity signal.

Table 2: Real-World Deepfake Attack Case Studies

Year	Attack Type	What Happened	Impact
2019	Audio deepfake fraud	Criminals used AI to clone the voice of a UK energy company CEO and called the subsidiary's director, instructing him to transfer money	£200,000 (\$243,000) was transferred to a criminal account
2020	Political disinformation	A deepfake video of Belgian Prime Minister Sophie Wilmès was circulated linking COVID-19 to environmental destruction	Widespread public confusion; the video was viewed millions of times before removal
2022	Live video impersonation	A deepfake video call impersonating Ukrainian President Zelensky was used to deceive several European city mayors	Multiple European leaders were manipulated during the active Russia-Ukraine conflict
2023	Corporate fraud	Fraudsters used real-time deepfake audio and video of a company CFO during a video call to trick an employee	HK\$200 million (approximately \$25 million USD) was transferred to criminal accounts
2024	Election interference	AI-generated fake audio of US President Biden circulated before a primary election, falsely telling voters not to vote	New Hampshire Department of Justice launched a criminal investigation

Table 2: Documented real-world deepfake attacks demonstrating the financial and political consequences of this technology.

3.5 AI-Based Detection Methods: Detailed Analysis

3.5.1 Spatial Domain Analysis Using Convolutional Neural Networks

The most widely deployed approach to deepfake detection involves training CNN-based classifiers to identify visual artefacts in individual video frames. These artefacts include unusual blending at face boundaries, irregular skin texture, unnatural reflections in the eyes (corneal inconsistencies), asymmetric facial muscle movements, and subtle colour banding [8].

CNN-based detectors are trained on large labelled datasets of real and fake images. Given sufficient training data that matches the type of fake being detected, they can achieve very high accuracy. The primary limitation is their inability to generalise: a CNN trained on FaceSwap-generated fakes performs poorly when tested against SimSwap or Stable Diffusion-generated content. Addressing this generalisation gap remains one of the most active areas of research in deepfake detection.

3.5.2 Temporal Consistency Analysis

Over time, human faces follow certain biological patterns. For example, blinking happens about every 4 to 6 seconds, lip movement must match speech phonetics exactly, head movement follows physical inertia, and facial muscle contractions follow anatomical limits. Algorithms that make deepfakes, especially those that work frame by frame, often make small mistakes in these time patterns [14].

To model these patterns over time, we use Long Short-Term Memory (LSTM) networks and other recurrent architectures. Temporal detectors can find inconsistencies that spatial detectors miss by looking at sequences of video frames instead of single frames. For example, they can find an unnatural transition between blinks or a brief desynchronization between a head turn and the predicted natural follow-through motion.

3.5.3 Frequency Domain Analysis

Each digitized photo possesses visible information at the pixel level and an intrinsic mathematical characteristic in the frequency domain. The characteristic presents an image based on the patterns contained within it and their frequencies. Studies have shown that GAN-produced images possess unique characteristics in the frequency domain. The upsampling process employed during generation results in patterns that manifest themselves as periodic peaks on the Fourier spectrum of an image [15].

3.5.4 Biometric Physiological Signal Analysis

One of the most cutting-edge ways to find things uses a basic biological fact: real human faces have living physiological signals that AI-generated faces can't copy. When blood flows through the tiny capillaries in your face with each heartbeat, it makes small, rhythmic changes in the color of your skin. These changes are so small that you can't see them with the naked eye, but AI systems that look at high-resolution video can measure them. The name of this method is **remote photoplethysmography (rPPG)** [16].

The IEEE Transactions on Pattern Analysis and Machine Intelligence published Intel's FakeCatcher system in 2023. It showed that rPPG-based detection could find deepfakes with about 96 percent accuracy. The system looks at several areas of the face at once to see if the blood flow signal is consistent across the face. In real video, the heart rhythm is the same in all parts of the face. In deepfakes, this coherent signal is either missing or physically impossible [16].

Table 3: Comparative Analysis of AI-Based Deepfake Detection Methods

Detection Method	What It Checks	How It Works (Simple)	Best Accuracy	Key Weakness
CNN Spatial Analysis	Individual video frames	Learns to spot tiny visual glitches in pixel patterns that AI faces tend to create	~95% on known fakes	Fails badly on unseen or new deepfake types — generalisation gap
LSTM Temporal Analysis	Video over time	Checks whether blinking, head movement, and lip sync remain natural throughout the whole video	~90%	Needs long video clips; slow to process
Vision Transformer (ViT)	Whole image at once	Uses attention across the entire face — catches lighting, shadow, and texture mismatches that CNNs miss	~97%	Needs large training datasets; high computing cost
Frequency Analysis (FFT)	Hidden image structure	Detects invisible AI fingerprint patterns left in image frequency data — like a hidden watermark from the AI	~88%	Fails when video is heavily compressed or re-uploaded to social media
Biometric rPPG	Biological signals	Detects whether a natural heartbeat pulse signal exists in facial skin colour — AI fakes cannot reproduce this	~85%	Needs high-quality video; advanced fakes can partially mimic the signal
Multimodal Fusion	Everything combined	Combines spatial, temporal, frequency, and biometric checks into one unified detection system	~98%	Very complex; high hardware requirements; slower processing speed

Table 3: Six detection methods compared by approach, accuracy, and key weaknesses.

4. Future Developments in the Area

The deepfake threat landscape will keep changing quickly. But a wide range of technical, organizational, and legal solutions are being worked on right now. Research literature and industry activity have identified six key future directions [17].

4.1 Blockchain-Based Content Provenance and Authentication

One of the most different ways to deal with the deepfake problem is to switch from reactive detection, which tries to find fakes after they are made, to proactive provenance, which checks the authenticity of content as soon as it is captured. This is possible thanks to blockchain technology [18].

The Coalition for Content Provenance and Authenticity (C2PA) is a group of companies, including Adobe, Microsoft, Intel, the BBC, and Reuters, that sets technical standards. They have come up with a way to add cryptographic signatures to media files when they are made. When a camera records video, it makes a unique cryptographic hash of the content and stores it on a blockchain ledger that can't be changed. Any change to the video after that, even one frame, makes the hash invalid, which is immediate and verifiable proof of tampering [18].

Some professional camera hardware and content management systems are already using this system. Its widespread use, especially as a required standard for news organizations, government communications, and legal proceedings, is likely to make deepfake content much less credible, even when detection algorithms fail.

4.2 Federated Learning for Privacy-Preserving Detector Training

There isn't enough information to make it easy to teach people how to use deepfake detection systems. To work well, these models need a lot of labeled data from both real and fake videos. People are very worried about their privacy, though, when it comes to some of the most useful information, like personal videos. Putting all of this private information in one place can cause big moral and legal problems [19].

This problem can be solved by federated learning, which keeps the data separate. Instead of sending private videos to a central server, each group trains its own version of the detection model using data from its own area. The model's parameters are numbers that show trends without giving away any personal information. Then, they are combined to make a better model for everyone [19].

Nguyen et al. (2022) discovered that a federated deepfake detection system, trained on data from various organizations, performed nearly as effectively as a centrally trained model, exhibiting only a 2–4% disparity in accuracy. Banks, social media sites, government agencies, and police can all work together to make detection systems better without putting people's private information at risk.

4.3 Real-Time On-Device Detection

Real-time detection is now one of the most important skills in the field because of the rise of live deepfake call fraud, in which criminals pretend to be executives or family members during video calls. The best detection models right now are too slow and need too much computer power to work with the latency limits of live video [20].

The answer is to combine two trends: the creation of ultra-lightweight neural network architectures that are optimized for inference speed (like MobileNet variants and efficient

transformer models) and the addition of Neural Processing Units (NPUs) to consumer devices. Apple, Qualcomm, and Google all make modern flagship smartphones that already have dedicated NPU hardware. This hardware can run AI inference in less than 10 milliseconds per frame, which is fast enough for real-time analysis of live video calls.

Several tech companies are working on SDKs for on-device deepfake detection that could be built right into videoconferencing platforms. These SDKs would alert users in real time if a suspected deepfake is found during a live call. This feature should be available for purchase between 2025 and 2027.

4.4 Adversarially Hardened Detection Systems

As deepfake detection tools have improved, a new adversarial dynamic has emerged: deepfake creators are now specifically designing their outputs to evade known detectors. By applying small, invisible perturbations to a deepfake — a technique drawn from the field of adversarial machine learning — attackers can cause detection accuracy to fall from over 95 percent to below 10 percent with no visible change to the human eye [21].

Two main countermeasures are being developed. First, adversarial training — intentionally including adversarially perturbed examples in the detector's training data so it learns to be resistant to such attacks. Second, ensemble detection — using multiple independent detectors simultaneously, each checking different aspects of the media. A perturbation that fools a spatial CNN is unlikely to simultaneously fool a temporal LSTM, a frequency-domain analyser, and a biometric rPPG detector. Requiring agreement across all four creates a system that is exponentially harder to defeat.

4.5 Universal AI Content Watermarking

In parallel with detection-focused approaches, a different strategy is gaining momentum: requiring AI tools themselves to embed invisible watermarks into all the content they generate. Unlike provenance systems that authenticate content at capture, AI watermarking embeds an invisible but detectable marker at the generation stage — meaning any content produced by a specific AI model carries an identifying signature that can be read by a separate watermark detector [17].

Major AI companies including Google DeepMind (with their SynthID system), Meta, and OpenAI have all begun experimenting with watermarking standards for AI-generated content. The challenge is ensuring watermarks survive common post-processing operations like cropping, compression, colour adjustment, and re-encoding — all of which a bad actor might apply to try to remove the watermark. Current research suggests that robust watermarking is achievable, but a universal industry standard has not yet been agreed upon.

4.6 Legal and Regulatory Frameworks

Many people are viewing legal responsibility, along with solutions, as one key way to stop deepfakes. The 2024 AI Act is the European Union's (EU) first big set of rules for AI. Deepfake technology and software posing risk, affecting both consumers and industry. According to it, AI-generated content requires a clear marking, and there are stringent rules for doing that [22]. The same applies for using tools that fake media.

The US's proposed DEEPFAKES Accountability Act will prohibit the making of deepfakes without additional disclosure, like a watermark. If an individual engages in deepfaking for the purposes of fraud, harassment, or altering election outcomes, they would face greater penalties. Several states, including California, Virginia and Texas, have now passed laws to prevent challenges like interfering with elections, sharing others' private content without their permission [22].

China has mandated all forms of AI content posted on any of its platforms must clearly state where it has come from. In Singapore, the deepfake misinformation was also dealt with using the Protection from Online Falsehoods and Manipulation Act (POFMA). As deepfakes grow increasingly prevalent—as well as dangerous—it is becoming more common for people to say that countries need to harmonize their laws, like in the case of cybercrime.

Table 4: Summary of Future Development Areas in Deepfake Detection and Prevention

Future Area	What It Is	Why It Helps	Expected By
Blockchain Provenance (C2PA)	A digital fingerprint is added to every video at the exact moment it is recorded. Any later edit breaks the fingerprint.	Makes it easy to prove a video has been tampered with — authentication at the source, not after the fact	2025–2027
Federated Learning	Multiple organisations train a shared deepfake detector without ever sharing raw videos with each other. Only the AI's learned parameters are shared.	Allows much larger, more diverse training data while fully protecting private and sensitive media	2026–2028
Real-Time On-Device Detection	Neural Processing Units (NPUs) built into smartphones and laptops run lightweight detection models fast enough to check live video calls as they happen	Stops criminals from using live deepfakes for impersonation during financial calls and legal meetings	2025–2027
Adversarially Hardened Detectors	Detection models are trained specifically against adversarial deepfakes — fakes that have been crafted to deliberately confuse detectors	Dramatically improves detector robustness against criminals who adapt their deepfakes to evade detection	Ongoing now

Universal Watermarking Standards	AI-generated content is automatically embedded with invisible watermarks by the AI tool itself at the moment of creation	Creates a global standard for identifying AI content — works even without a separate detector	2026–2028
Legal & Regulatory Frameworks	Laws such as the EU AI Act (2024) and the US DEEPFAKES Accountability Act set mandatory requirements for labelling, disclosure, and criminal penalties	Provides legal accountability alongside technical solutions — bad actors face real-world consequences	2026–2030

Table 4: Six future development areas with descriptions, benefits, and expected timelines.

5. Conclusion

The advancement of deepfake technology in the last decade has gone from a face swap tool to an AI-generated media system capable of producing highly convincing synthetic video, audio and imagery at scale. The progress has posed great dangers to cybersecurity, democracy and to individual privacy among others.

The deepfake timeline detailed in this report runs through 2017 GAN-based face-swap tools and all the way through the foundation model period of 2023 and after. It examined four cutting-edge AI detection techniques – CNN spatial analysis, LSTM temporal analysis, frequency-domain detection and biometric rPPG signal analysis. Currently, multimodal fusion systems with all-four have the highest detection accuracy at about 98 percent according to this analysis. The continuing arms race facilitated by advancements in generation technologies was also cited in the report that is presently outpacing current detection capabilities in use.

The following six developments are likely to be particularly relevant in the future: blockchain-based content provenance, federated learning for privacy-preserving detector training, real-time on-device detection, adversarially hardened detection systems, universal AI watermarking standards and improved and strengthened legal and regulatory frameworks, e.g. EU AI Act (2024). Nothing will work alone to get the job done. Tackling deepfakes’ threat will require simultaneous and coordinated progress in technical, institutional and legislative spaces. This will not be achieved by one measure alone. Moving ahead, tackling the detection models’ generalisation gap, formulating internationally recognised watermarking standards and assessing the viability of new regulatory systems in real life should be the focus of future work.

Deepfake tech is more complex and evolving than most threats in the cybersecurity world today. According to the report, which comes amid rising climate activism, a joint outcome of informed, managed deployment of carbon dioxide removal technologies is vital for the global climate...continue reading.

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